

Line Integrals

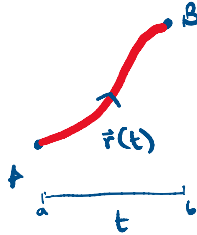
Monday, May 19, 2025 11:00 PM

Slide: Where have you seen line integrals before?

12 - Line Integrals

12.1 There are a few different forms of line integral. They all share the same ideas:

- Some curve C is parametrized by $\vec{r}(t)$, $a \leq t \leq b$,
- C is oriented such that there is a starting point $A = \vec{r}(a)$ and an end-point $B = \vec{r}(b)$
- Some quantity (could be scalar field f or vector field $\vec{F}$) is integrated along the curve.



We will need to be clear on the form that is being used.

12.2 First form:

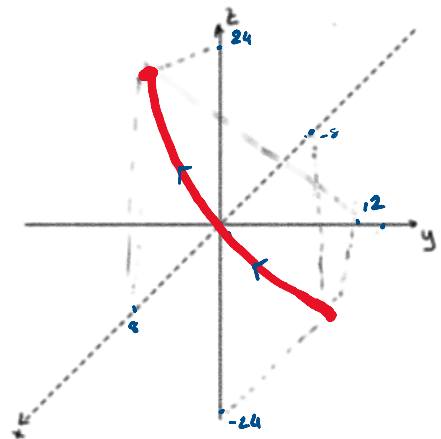
For a scalar function $f(x, y, z)$ and curve C , compute

$$(8^*) \quad \int_C f(\mathbf{r}) dt = \int_a^b f(\mathbf{r}(t)) dt$$

Usage: friction along a curve, internal resistance in a wire, temperature/thermal energy along a rod, cost of materials to build a fence, etc.

12.3 Example. Let $f = xyz$, $C: \vec{r} = [4t, 3t^2, 12t]$, $-2 \leq t \leq 2$.

$$\begin{aligned} \int_C f dt &= \int_{-2}^2 f(4t, 3t^2, 12t) dt \\ &= \int_{-2}^2 x(t) y(t) z(t) dt \\ &= \int_{-2}^2 (4t)(3t^2)(12t) dt \\ &= \int_{-2}^2 144 t^4 dt \\ &= 144 \frac{t^5}{5} \Big|_{-2}^2 \end{aligned}$$



$$\begin{aligned}
 & \left. \frac{1}{5} x^5 \right|_{-2}^2 \\
 &= \frac{144}{5} \left(2^5 - (-2)^5 \right) \\
 &= \frac{144}{5} (32 - (-32)) \\
 &= \frac{144}{5} (32 + 32) = \frac{144}{5} (64) = \frac{9216}{5} = 1843.2
 \end{aligned}$$

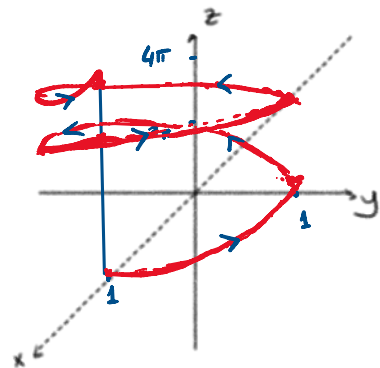
12.4 length of a curve: $l = \int_a^b \sqrt{\vec{r}' \cdot \vec{r}'} dt$ is also a line integral of this type,
 where $f = \sqrt{\vec{r}' \cdot \vec{r}'} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}$ is the speed of parametrization.

e.g. find the length of $C: \vec{r} = [\cos t, \sin t, t]$, $0 \leq t \leq 4\pi$

Soln:

$$\begin{aligned}
 f &= \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} \\
 &= \sqrt{(-\sin t)^2 + (\cos t)^2 + (1)^2} \\
 &= \sqrt{\sin^2 t + \cos^2 t + 1} = \sqrt{1+1} = \sqrt{2}
 \end{aligned}$$

$$\text{The } \int_C f dt = \int_0^{4\pi} \sqrt{2} dt = \sqrt{2} (4\pi) = 4\sqrt{2} \pi$$



12.5 Note that

$$\underbrace{\sqrt{\vec{r}' \cdot \vec{r}'}}_{\text{speed}} \underbrace{dt}_{\text{time}} = \underbrace{dl}_{\text{distance or length}}$$

Second Form of Line Integral

$$\int_C f(\vec{r}) dl = \int_a^b f(\vec{r}(t)) \underbrace{\sqrt{\vec{r}' \cdot \vec{r}'}}_{\text{extra factor of speed}} dt$$

Usage: Calculating total mass from linear mass density $\mu(\vec{r}(t))$,
 total charge from linear charge density $\sigma(\vec{r}(t))$.

- Remarks - This is often a very difficult integral to compute exactly.
- We will ignore this form in this course.
 - This form is actually more common in Engineering than the first form.

12.6 Line Integrals of vector fields

Third form:

A **line integral** of a vector function $\mathbf{F}(\mathbf{r})$ over a curve $C: \mathbf{r}(t)$ is defined by

$$(3) \quad \int_C \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt \quad \mathbf{r}' = \frac{d\mathbf{r}}{dt}$$

Example Let $\vec{F} = \begin{bmatrix} y^2 \\ -x^2 \end{bmatrix}$ and $C: y = 4x^2$ from $(0,0)$ to $(1,4)$

Find $\int_C \vec{F}(\vec{r}) \cdot d\vec{r}$

Slide: How is C parametrized?

Soln Start by parametrizing $C: \begin{matrix} x(t) = t \\ y(t) = 4t^2 \end{matrix}, \quad 0 \leq t \leq 1$

Then $x'(t) = 1, y'(t) = 8t$

$$\text{So } \frac{d\vec{r}}{dt} = \begin{bmatrix} x'(t) \\ y'(t) \end{bmatrix} = \begin{bmatrix} 1 \\ 8t \end{bmatrix}$$

$$\text{Similarly: } \vec{F} = \begin{bmatrix} y^2 \\ -x^2 \end{bmatrix} \Rightarrow \vec{F}(\vec{r}(t)) = \begin{bmatrix} (4t^2)^2 \\ -(t)^2 \end{bmatrix} = \begin{bmatrix} 16t^4 \\ -t^2 \end{bmatrix}$$

$$\text{Then } \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) = \begin{bmatrix} 16t^4 \\ -t^2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 8t \end{bmatrix} = 16t^4 - 8t^3$$

$$\text{So } \int_C \vec{F} \cdot d\vec{r} = \int_0^1 \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) dt = \int_0^1 (16t^4 - 8t^3) dt$$

$$= \left[\frac{16t^5}{5} - \frac{8t^4}{4} \right]_0^1$$

$$= \frac{16}{5} - \frac{8}{4} = \frac{16}{5} - 2 = \frac{16-10}{5} = \frac{6}{5} = 1.2$$

12.7 This form: $\int_C \vec{F} \cdot d\vec{r}$ is usually considered as the "work done" by the force $\vec{F}$ in moving an object along the curve C .

In the ideal case where the entirety of the force applied just changes the momentum of a object with mass m , we get that this work done gets converted to kinetic energy.

Slide: Newton's Second Law of Motion

EXAMPLE 4 Work Done Equals the Gain in Kinetic Energy

Let $\mathbf{F}$ be a force, so that (3) is work. Let t be time, so that $d\mathbf{r}/dt = \mathbf{v}$, velocity. Then we can write (3) as

$$(6) \quad W = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{v}(t) dt.$$

Now by Newton's second law, that is, force = mass $\times$ acceleration, we get

$$\mathbf{F} = m\mathbf{r}''(t) = m\mathbf{v}'(t),$$

where m is the mass of the body displaced. Substitution into (5) gives [see (11), Sec. 9.4]

$$W = \int_a^b m\mathbf{v}' \cdot \mathbf{v} dt = \int_a^b m \left(\frac{\mathbf{v} \cdot \mathbf{v}}{2} \right)' dt = \frac{m}{2} |\mathbf{v}|^2 \Big|_{t=a}^{t=b}.$$

On the right, $m|\mathbf{v}|^2/2$ is the kinetic energy. Hence *the work done equals the gain in kinetic energy*. This is a basic law in mechanics. ■

12.8 Since $\int_C \vec{F} \cdot d\vec{r}$ is a scalar, it should not depend on the parametrization $\vec{r}(t)$

THEOREM 1

Direction-Preserving Parametric Transformations

Any representations of C that give the same positive direction on C also yield the same value of the line integral (3).

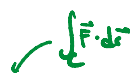
PROOF The proof follows by the chain rule. Let $\mathbf{r}(t)$ be the given representation with $a \leq t \leq b$ as in (3). Consider the transformation $t = \phi(t^*)$ which transforms the t interval to $a^* \leq t^* \leq b^*$ and has a positive derivative dt/dt^* . We write $\mathbf{r}(t) = \mathbf{r}(\phi(t^*)) = \mathbf{r}^*(t^*)$. Then $dt = (dt/dt^*) dt^*$ and

$$\begin{aligned} \int_C \mathbf{F}(\mathbf{r}^*) \cdot d\mathbf{r}^* &= \int_{a^*}^{b^*} \mathbf{F}(\mathbf{r}(\phi(t^*))) \cdot \frac{d\mathbf{r}}{dt} \frac{dt}{dt^*} dt^* \\ &= \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \frac{d\mathbf{r}}{dt} dt = \int_C \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r}. \end{aligned} \quad \blacksquare$$

12.9 Path Dependence

THEOREM 2

Path Dependence



THEOREM 2

Path Dependence



The line integral (3) generally depends not only on $\mathbf{F}$ and on the endpoints A and B of the path, but also on the path itself along which the integral is taken.

Example Let $\vec{F} = \begin{bmatrix} x^2y \\ -4xy^2 \\ 8z^2x \end{bmatrix}$. Consider two paths from $(0,0,0)$ to $(1,1,1)$

$$C_1: \vec{r}_1(t) = [t, t, t], \quad 0 \leq t \leq 1$$

$$C_2: \vec{r}_2(t) = [t, t^2, t^3], \quad 0 \leq t \leq 1$$

$$\text{Find } \int_{C_1} \vec{F} \cdot d\vec{r} \quad \text{and} \quad \int_{C_2} \vec{F} \cdot d\vec{r}$$

Soln: For C_1 : $\vec{r}_1'(t) = [1, 1, 1]$

$$\begin{aligned} \text{So } \vec{F} \cdot \vec{r}'(t) &= \begin{bmatrix} x^2y \\ -4xy^2 \\ 8z^2x \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = x^2y - 4xy^2 + 8z^2x \\ &= (t^2(t) - 4(t)(t)^2 + 8(t)^2(t)) \\ &= t^3 - 4t^3 + 8t^3 = 5t^3 \end{aligned}$$

$$\text{Then } \int_{C_1} \vec{F} \cdot d\vec{r} = \int_0^1 5t^3 dt = \frac{5}{4} = 1.25$$

For C_2 : $\vec{r}_2'(t) = [1, 2t, 3t^2]$

$$\vec{F}(\vec{r}(t)) = \begin{bmatrix} x^2y \\ -4xy^2 \\ 8z^2x \end{bmatrix} = \begin{bmatrix} (t)^2 \cdot (t^2) \\ -4(t)(t^2)^2 \\ 8(t^3)^2(t) \end{bmatrix} = \begin{bmatrix} t^4 \\ -4t^5 \\ 8t^7 \end{bmatrix}$$

$$\text{So } \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) = \begin{bmatrix} t^4 \\ -4t^5 \\ 8t^7 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2t \\ 3t^2 \end{bmatrix} = t^4 - 8t^6 + 24t^9$$

$$\text{Finally: } \int_{C_2} \vec{F} \cdot d\vec{r} = \int_0^1 (t^4 - 8t^6 + 24t^9) dt = \left[\frac{t^5}{5} - \frac{8t^7}{7} + \frac{24t^{10}}{10} \right]_0^1$$

$$= \frac{1}{5} - \frac{8}{7} + \frac{24}{10} = \frac{14 - 80 + 168}{70} = \frac{102}{70} \approx 1.46$$

The results are very different!

The results are very different!

12.10 However, we do know that Work done is often path independent!?

e.g. Work done under gravity only cares about change in height, not the path taken to get that change.

12.11 Path-Independence: Foreshadow

THEOREM

Path Independence

A line integral (1) with continuous F_1, F_2, F_3 in a domain D in space is path independent in D if and only if $\mathbf{F} = [F_1, F_2, F_3]$ is the gradient of some function f in D ,

$$(2) \quad \mathbf{F} = \text{grad } f, \quad \text{thus,} \quad F_1 = \frac{\partial f}{\partial x}, \quad F_2 = \frac{\partial f}{\partial y}, \quad F_3 = \frac{\partial f}{\partial z}.$$

How would we know whether $\vec{F} = \vec{\nabla} f$ for some scalar f ? Stay tuned for next week

12.14 Line integral of $\vec{F}$
Fourth form

$$\int_C \mathbf{F}(\mathbf{r}) dt = \int_a^b \mathbf{F}(\mathbf{r}(t)) dt = \int_a^b [F_1(\mathbf{r}(t)), F_2(\mathbf{r}(t)), F_3(\mathbf{r}(t))] dt.$$

Note: Here the line integral gives a vector.

Usage: Rare, almost never. Sometimes with time-evolution of fields.

Remark: We will mostly ignore this form.

e.g. $\vec{F} = \begin{bmatrix} y^{1/3} \\ x^{1/3} \\ 0 \end{bmatrix}$, C : the hypocycloid $\vec{r} = [\cos^3(t), \sin^3(t), 0]$,
 $0 \leq t \leq \pi/4$

$$\begin{aligned} \text{Then} \quad \int_C \vec{F}(\vec{r}) dt &= \int_0^{\pi/4} \begin{bmatrix} (\sin^3(t))^{1/3} \\ (\cos^3(t))^{1/3} \\ 0 \end{bmatrix} dt = \begin{bmatrix} \int_0^{\pi/4} \sin t dt \\ \int_0^{\pi/4} \cos t dt \\ \int_0^{\pi/4} 0 dt \end{bmatrix} = \begin{bmatrix} -\cos t \big|_0^{\pi/4} \\ \sin t \big|_0^{\pi/4} \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} -(\cos(\pi/4)) - (-\cos(0)) \\ -1/\sqrt{2} + 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1/\sqrt{2} + 1 \\ 1 - 1/\sqrt{2} \\ 0 \end{bmatrix} \end{aligned}$$

$$= \begin{bmatrix} (-\cos(\pi/4)) - (-\cos(0)) \\ (\sin(\pi/4) - \sin(0)) \\ 0 \end{bmatrix} = \begin{bmatrix} -1/\sqrt{2} + 1 \\ 1/\sqrt{2} - 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 - 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}.$$